MILITARY SPECIFICATION

VALVE, ANTIBACK DRAFT, PROTECTIVE SHELTER, M2

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of valve for protective shelters.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-571 — Bolts; Nuts; Studs; and Tap-Rivets (and Material for Same).

NN-P-515 — Plywood, Container Grade.

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

LLL-F-291 — Fiberboard; Corrugated, Single-Face, (Flexible).

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape, Pressure-Sensitive Adhesive, Waterproof — For Packaging and Sealing.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-121 — Barrier Material, Greaseproofed, Flexible (Waterproofed).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

CHEMICAL CORPS

D5-65-31 — Valve, Antiback Draft, Protective Shelter, M2—Bill of Material.

D5-65-36 — Cells, Packing.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publication forms a part of this specification. Unless otherwise indicated. the issue in effect on date of invitation for bids shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee. 202 Chicago Union Station. Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials cited on Drawings D5–65–31, or on subsidiary drawings, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 *Components.* All components of the item shall conform to the specifications and drawings listed on Drawing D5–65–31 and subsidiary drawings.

3.2 **Manufacture and assembly.** The valve shall be manufactured and assembled as specified on Drawing D5–65–32 and its subsidiaries, which are listed on Drawing D5–65–31.

3.3 **Closure.** When the valve is tested as specified in 4.5 the gap between valve cover (Dwg. B5–65–40) and gasket (Dwg. B5–65–11) shall be not more than $\frac{1}{32}$ inch.

3.4 **Workmanship.** The valve shall be free from contamination (foreign matter) and damage such as dents, cracks, or deformations which would impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Supplier's responsibility.** Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Objective evidence.** The supplier shall present objective evidence as required by the Government representative that all material and components are in accordance with 3.1 and section 5 (see 6.5).

4.3 **Alternate inspection (including testing) procedures.** The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of the specification will govern.

4.4 Inspection provisions.

4.4.1 *Regular production lot.* A regular production lot shall consist of the valves produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions.

4.4.2 *Sampling.* Sampling shall be conducted in accordance with Standard MIL–STD–105. The level and degree of inspection will be as specified in the contract.

4.4.3 *Examination.* Sample valves from the regular production lot shall be examined in accordance with the classification of defects and the AQL's as designated by this specification and with Standard MIL–STD–105. Sample items will be tested as specified in 4.5.

4.4.4 *Classification of defects.*

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4.4.4.1 *Valve, antiback draft, M2 — assembly (Dwg. D5-65-32).*

Categories and defects	Acceptance standards
Major A — AQL 0.25 percent defective	
1a. Valve closure incorrect.	4.5
Major — — AQL 1.0 percent defective	
1. Component missing, incorrect, or incorrectly assembled.	
2. Welding incorrect.	
3. Counterweight not free to move the entire length of shaft.	
4. Damage (see 3.4).	
Minor — AQL 2.5 percent defective	
51. Protective finish missing, inadequate, or incorrect.	
52. Glue missing or inadequate.	
Incidental — AQL 10.0 percent defective	
101. Contamination (see 3.4).	

4.4.4.2 *Preparation for delivery* (sec. 5).

Categories and defects
Major — AQL 4.0 percent defective
1. Contamination on unpainted surfaces.
2. Preservative incorrect or missing, where required.
3. Plywood sheet missing or incorrectly applied.
4. Container closures incorrect.
5. Containers incorrect.
6. Loose package or pack.
7. Fiberboard cells missing or incorrectly applied.
8. Strapping loose or missing.
9. Containers torn or broken.
10. Marking incorrect, missing, or illegible.
Minor — AQL 6.5 pecent defective
51. Exterior box sleeve missing.

4.5 Closure test. With the valve in operating position (i.e., as if mounted in a vertical wall with hinges at top), and with the counterweight so adjusted that the valve cover is closed, the widest gap (if any) between cover and gasket shall be measured.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. Unpainted surfaces of the valve (including threaded areas) shall be cleaned and dried in accordance with Specification MIL–P–116, using cleaning method C-3 and drying method D-1 or D–4.

5.2 Preserving and packaging.

5.2.1 *Level A.* The unpainted surfaces, cleaned and dried as specified in 5.1, shall be coated with type P–2 preservative of Specification MIL–P–116 (see 6.4).

5.2.1.1 *Counterweight assembly.* The preserved counterweight assembly shall be wrapped in barrier material conforming to type II, grade A, class 2, of Specification MIL–B–121, securing the wrap in place with gummed tape, conforming to Specification UU–T–111. Fiberboard conforming to type II of Specification LLL–F–291 shall be wound around the portions of the processed shaft projecting beyond the counterweight, forming cylinders measuring 3¼ to 3 5/16 inches in length and 2¼ to 2 5/16 inches in outside diameter. The fiberboard wraps shall be secured in place with the same type of gummed tape as cited above. The assembly shall be packaged in a box conforming to W5c, style RSC, of Specification PPP–B–636 and measuring 9⅛ by 2¼ by 2¼ inches inside. The box shall be closed in accordance with the appendix to its specification for interior boxes.

5.2.1.2 *Valve assembly.* The adjustable flange shall be bolted to the valve body near the hinge. The four corners of the flange shall then be bolted to a rectangular piece of 1½ inch thick plywood measuring about 17¼ by 12¼ inches, with the plywood notched to accommodate the cover hinges. The plywood shall conform to type II of Specification NN–P–515. Cadmium or zinc-coated steel carriage bolts conforming to Specification FF–B–571 shall be used.

5.2.2 *Level B.* Not applicable.

5.2.3 *Level C.* The valve shall be cleaned, dried, and preserved as specified in 5.1 and 5.2.1, and then packaged in accordance with the manufacturer's commercial practice.

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5.3 **Packing.**

5.3.1 *Level A.* The counterweight and valve assemblies, packaged as specified in 5.2.1, shall be packed in a box conforming to V2s or V3s, style RSC–SL, of Specification PPP–B–636 and measuring 17¼ to 17$^{5}/_{16}$ by 12¼ to 12$^{5}/_{16}$ by 11½ to 11$^{9}/_{16}$ inches inside. The assemblies shall be blocked as shown in figure 1, using cells conforming to Drawing D5–65–36. The box shall be closed and strapped in accordance with the appendix to the box specification. In addition all seams including the manufacturer's joint of the box shall be sealed with minimum 3 inch wide tape conforming to type III, class 1, of Specification PPP–T–60, with the tape being extended at least 3 inches over all corners and edges of the box.

5.3.2 *Level B.* Not applicable.

5.3.3 *Level C.* The valve, packaged as specified in 5.2.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.4 **Marking.** In addition to any special marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The valve is intended to automatically regulate the pressure and airflow rate between the outer air locks of protective shelters and the atmosphere.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) The title, number, and date of this specification.

(b) The level or levels of protection required (see sec. 5).

6.3 **Supersession data.** This specification supersedes Chemical Corps Purchase Description 197–54–618, dated 7 October 1955.

6.4 **Preservative.** Type P–2 preservative (Specification MIL–P–116) is grade 2, soft film compound, conforming to Specification MIL–C–16173.

6.5 **Objective evidence.** Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.6 **Pilot lot.** A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Chemical Corps

Navy—Bureau of Yards and Docks

Air Force

Preparing activity:

Army—Chemical Corps

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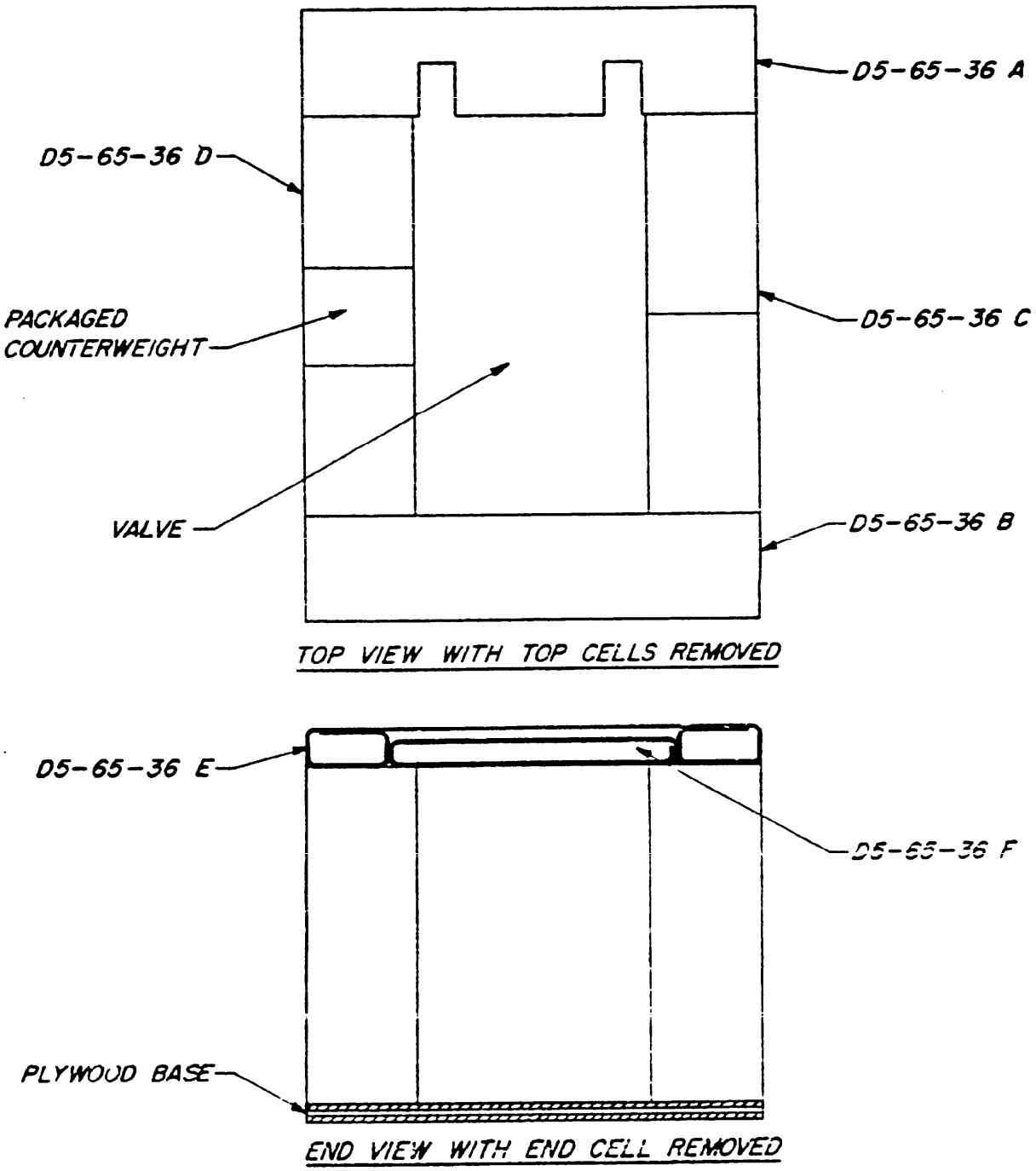


FIGURE 1. *Packing, EIR6 antiback draft valve.*

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